

# ADI KARTHIKEYA SUNKARA BALASUBRAMANYAM

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

## PROFESSIONAL SUMMARY

AI/ML Engineer with 3+ years of experience building machine learning, predictive analytics, and Generative AI solutions across retail, supply chain, and enterprise domains. Skilled in designing scalable ML pipelines, recommendation systems, and agentic workflows. Strong foundation in Python, AWS, Spark, and modern AI frameworks to drive tangible business outcomes.

## SKILLS

**AI & ML:** Scikit-learn, TensorFlow, PyTorch, Deep Learning, RAG, LLMs (Claude, Gemini), Agentic Workflows

**Languages & Data:** Python, SQL, PySpark, R, Pandas, NumPy, Matplotlib

**Cloud & MLOps:** AWS, GCP, Docker, Kubernetes, MLflow, CI/CD

**Data Engineering & DBs:** Spark, Databricks, ETL, PostgreSQL, MongoDB, Snowflake, Pinecone, ChromaDB, Neo4j

## EXPERIENCE

### ML Engineer

Aug 2025 – Present

Walmart

Houston, TX

- Developed and deployed Machine Learning solutions for Demand Forecasting and Inventory Optimization, significantly improving operational efficiency across large-scale retail environments.
- Built recommendation and ranking systems leveraging customer clickstream and transaction data to enhance product discovery and conversion rates.
- Designed and productionized end-to-end ML pipelines covering data ingestion, feature engineering, and model deployment for both batch and real-time inference.
- Engineered scalable feature stores to ensure consistency between offline model training and online serving.
- Processed large-scale datasets including POS, pricing, and logistics data using Apache Spark.
- Implemented real-time model monitoring and automated retraining workflows to ensure lifecycle reliability.

### Data Scientist

May 2021 – Jun 2024

Reputable Technologies Pvt. Ltd.

India

- Partnered with product and supply chain teams to translate business requirements into production ML solutions.
- Built and deployed classification, clustering, and forecasting models using Python-based frameworks.
- Developed customer segmentation models using purchase history to enable targeted marketing campaigns.
- Designed data pipelines (ETL) for structured and unstructured data from IoT and enterprise systems.
- Optimized feature selection and model tuning pipelines to reduce training latency for large-scale datasets.

## PROJECTS

### RareDx, Agentic Rare Disease Diagnostic Assistant | LangGraph, Neo4j, ChromaDB, Groq

Mar 2026 – May 2026

- Engineered an AI diagnostic workflow in LangGraph to identify rare diseases, enriching ChromaDB with full Orphanet disease profiles using BioLORD embeddings.
- Lifted Top-1 diagnostic accuracy from 10% to 92.5% and Recall@5 to 100% across 40 adversarial benchmark cases by fixing critical semantic retrieval failures.
- Designed ablation experiments across 7 retrieval strategies, achieving a strictly calibrated zero-hallucination rate.

### ShopperAI, Agentic AI Shopping Assistant | Python, Google ADK, Gemini, LangChain

Dec 2025

- Built a smart, agent-driven shopping assistant using the Google ADK framework to guide users through complex purchasing decisions, including virtual try-ons.
- Designed deterministic, stage-based agent workflows to ensure reliable product recommendations and reduce user decision fatigue.
- Earned 4th place globally out of ~200 teams at the ODSC AI & Google Cloud Agentic AI Hackathon.

### CropGuard AI, Crop Disease Diagnosis | Next.js, Groq (LLM), TF-IDF, Open-Meteo

2026

- Created a free, symptom-based diagnostic tool for farmers that provides grounded treatment plans utilizing live weather data and localized climate history.
- Eliminated AI hallucinations by building a TF-IDF retrieval system that forces the LLM to pull advice strictly from a curated knowledge base of 31 verified diseases.

### Predictive Match ML Dashboard | Python, XGBoost, LightGBM, SHAP, Streamlit, NLP

Jun 2026

- Developed a full machine learning pipeline analyzing 60,000 profiles, engineering 52 features across bio sentiment (VADER) and lifestyle signals to predict match success.
- Trained and stacked ensemble models, leveraging SHAP explainability to deploy an interactive Streamlit dashboard that ranks personalized profile optimizations for users.

## EDUCATION

### University of Houston

Houston, TX

M.S. Data Science, GPA 3.9/4.0

May 2026

### Amrita Vishwa Vidyapeetham

Bengaluru, India

B.Tech, Electronics and Communication Engineering, GPA 3.50/4.00

May 2024